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Studierichting: Informatica

Oefeningenreeks 4A nr. 37.2

$$\int \frac{1}{(x-2)^2 + 9} dx = \int \frac{1}{(x-2)^2 + 3^2} dx$$

$$\text{Stel } t = x - 2 \Rightarrow dt = dx$$

$$= \int \frac{1}{t^2 + 3^2} dt = \int \frac{1}{3^2 \left(\left(\frac{t}{3} \right)^2 + 1^2 \right)} dt$$

$$\text{Stel } u = \frac{t}{3} \Rightarrow du = \frac{1}{3} dt \Rightarrow dt = 3du$$

$$= \int \frac{3}{3^2(u^2 + 1^2)} du = \frac{1}{3} \int \frac{1}{u^2 + 1} du$$

$$= \frac{1}{3} \text{Bgtan}(u) + c = \frac{1}{3} \text{Bgtan} \left(\frac{t}{3} \right) + c$$

$$= \frac{1}{3} \text{Bgtan} \left(\frac{x-2}{3} \right) + c$$

References